

# SANKUR KUNDU

## Software Engineer | Backend, Systems & Full-Stack

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## SUMMARY

Software engineer (B.Tech CSE, 2026) who builds systems from scratch and proves them correct: **tessel** (GPU kernels at **cuBLAS parity**), **tandem** (distributed training **bit-identical to PyTorch**), **quorumbd** (TLA+-verified SQL) and **faultline** (tests of **etcd**). Co-founded Gigzen as its sole engineer.

## PROJECTS

### **tessel** — Rust, zero dependencies: a tile language and compiler for GPUs and TPUs

- Designed a **Triton-style tile language** and wrote its compiler from scratch: typed tile IR, **linear layouts**, **CUDA for tensor cores** (ldmatrix, mma.sync, swizzled shared memory, pipelining), an **auto-tuner**, plus **Metal** and **TPU (Pallas)** backends.
- On an **NVIDIA T4**: GEMM at **parity with cuBLAS**, attention **1.3–1.9× faster than PyTorch SDPA**; an **LLM engine** on its kernels matches **vLLM's** output token for token at **1.05× its throughput** (32 sequences).

### **tandem** — Python, no torch.distributed or NCCL: distributed training from scratch

- Built the parallelisms behind large-model training on its own **ring all-reduce** over **CUDA IPC**: **DDP** with gradient bucketing, **ZeRO-1/2/3 (FSDP-style sharding)**, **GPipe** and **1F1B** pipelines, and a communication **profiler**.
- On **2× T4**, **DDP** and **ZeRO-1/2/3** train **bit-identically to PyTorch DDP/FSDP** on **NCCL**; all-reduce **within 2% of NCCL's bandwidth** at 64–256 MB; on an 86M-parameter GPT, **ZeRO-3** cuts per-GPU memory **17%**, **1F1B 32–52%** vs GPipe.

### **quorumbd** — Rust, zero dependencies: a distributed SQL database, PostgreSQL wire-compatible

- Built from scratch a **distributed SQL database** that **psql** connects to: **LSM storage**, **Raft**, a self-rebalancing **multi-Raft** store, **serializable ACID transactions** (Percolator), a SQL planner; **1,500 tpmC** on a **TPC-C**-derived benchmark.
- Proved by **deterministic simulation** (**10,000 Raft clusters**, **610K crashes**, histories machine-checked) and a **TLA+** model (**337K states**); found **6 real bugs**, catches all **15 planted**; **time-travel debugger** rewinds runs by replay.

### **faultline** — Rust, zero dependencies: Jepsen-style fault injection for etcd

- Runs real **etcd** clusters in **Linux network namespaces**, injects **iptables partitions**, **kill -9** crashes and **SIGSTOP** pauses, and checks histories for **linearizability**, **watch completeness** and **lost updates**.
- Tested **4 etcd release lines** with **1.26M operations** under **288 faults**: linearizability and watches held; traced **lost updates** under etcd locks to a paused leader applying a timed-out write **6 s** late, and showed **fenced writes** prevent it.

### Also built: **kiln** (an ML compiler) • **ferrolm** (an LLM server)

- kiln** compiles **ONNX** to **SIMD** and **CUDA**, **faster than torch.compile**, **ONNX Runtime** and **JAX/XLA** on CPU (**BERT 117 ms** vs 149–189); **ferrolm** serves LLMs with a paged KV cache and continuous batching (**3.4× throughput**).

## OPEN SOURCE

PR submitted to **PyTorch** (under review): [pytorch/pytorch#199441](https://github.com/pytorch/pytorch/pull/199441) removes **13 stale torch.compile (Inductor) test skips** (most cited a bug fixed in 2024) after checking each test now passes on **CPU** and **2× T4 GPUs**.

## EXPERIENCE

### Gigzen — Co-founder & Founding Engineer

Apr 2026 – Present

Bhubaneswar, India • Early-stage startup (incorporation in progress) • Sole engineer across every product

- Sole engineer** on **Waggle**, a delivery platform of **three Android apps** (riders, shops, customers) from one **TypeScript / React / Capacitor** codebase on **Supabase PostgreSQL**: **53 tables**, **90 RLS policies**, **154 SQL functions**.
- Put the business rules in the database, where no client can bypass them: **first-accept-wins dispatch** (45-second offer window), **4-digit handover codes**, **Razorpay/UPI** payments reconciled by **UTR**, **GST invoices** and **pg\_cron** billing runs.
- Security-tested** it with **150 automated attacks** on its database access rules (all blocked) and a **42-step Playwright** run.
- Built **Populace**, an open-source multi-user testing tool (**Node.js**, an **npm CLI**, **GitHub Action** and **Electron app**); its **first run** found **5 defects in 3.5 minutes** after manual testing had passed, one breaking **every signup**.

## SKILLS

**CS Fundamentals**: Data Structures & Algorithms, System Design, Distributed Systems, DBMS internals, OS, Networks

**Languages**: Rust, TypeScript, JavaScript, Python, SQL (PL/pgSQL), Java

**Backend & Data**: PostgreSQL, Supabase, Row-Level Security, **pg\_cron**, REST APIs, Node.js, FastAPI, Flask, Redis, Razorpay/UPI

**Frontend & Mobile**: React, Capacitor (Android), Electron, Leaflet/MapLibre maps, HTML, CSS

**Systems & performance**: CUDA (tensor cores), distributed training (DDP, ZeRO/FSDP), kernel compilers, SIMD, LLM serving

**Testing, DevOps & Cloud**: Simulation testing, fault injection, TLA+, Playwright, GitHub Actions, Docker, Kubernetes, GCP

## EDUCATION

B.Tech, Computer Science and Engineering (AI & ML) — IEM Kolkata

Sep 2022 – Apr 2026